MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL,
50 OHMS, M17/205-00018 and M17/205-00050

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-17.

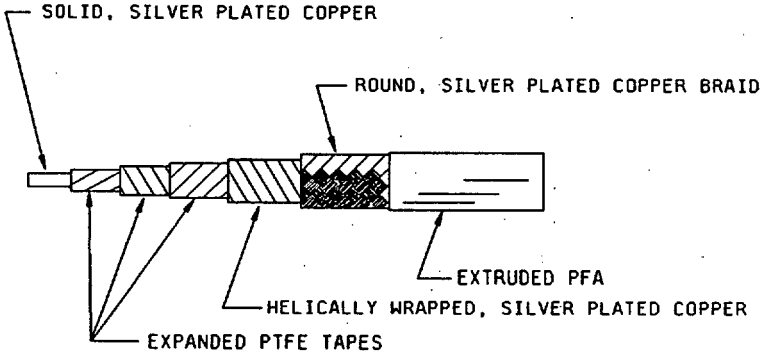


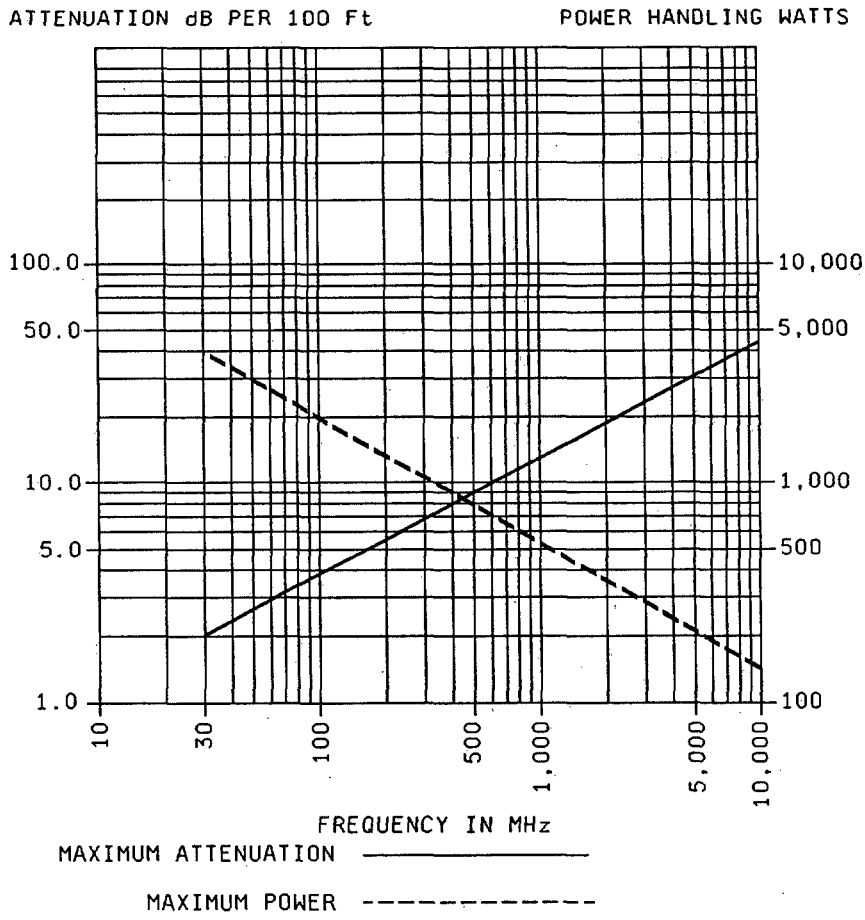
FIGURE 1. General configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid silver coated copper wire, Overall diameter: .0298 inch ±.0002
Dielectric core	Type F-6: Expanded PTFE tape. Diameter: .083 inch ±.003. 50% overlap minimum, 66% nominal.
Outer conductor Inner shield helical wrap	Double shield: Helical wrap plus braid Diameter: .108 inch ±.002. Carriers: N/A Ends: N/A Picks/inch: N/A Wire type: Flat, silver plated copper wire. .060 inch ±.002 x .002 inch ±.0003. 45% overlap minimum.
Outer braid	Carriers: 16 Ends: 5 Picks/inch: 25 ±2 Wire type: AWG 38, silver plated. Diameter: .108 ±.002, coverage 90%, minimum.
Jacket	Type XIII extruded PFA Diameter: .120 inch +.003.

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ATTENUATION	
MHz	dB
50	2.9
400	8.2
3,000	22.9
11,000	45.1

FIGURE 2. Power rating and attenuation at 25°C sea level.

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ENGINEERING INFORMATION:

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 50 GHz, maximum.

Velocity of propagation: 82 percent, nominal.

Power ratings: See figure 2.

Operating temperature range: -55°C to +200°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 12 ohms per 1000 feet, maximum.

Elongation: 25 percent, minimum.

Engineering notes: This cable is useful in general purpose, low loss, high temperature applications (connector series 2.4 mm, 3.5 mm and SMK).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: Not applicable.

Adhesion of conductors:

Inner conductor to core: Not applicable.

Aging stability: Not applicable.

Cold bend: -55°C ±2°C.

Dimensional stability: +200°C ±5°C.

Inner conductor from core: .187 inch, maximum.

Inner conductor from jacket: .312 inch, maximum.

Contamination: Not applicable.

Flame propagation: Applicable.

Acid gas generation: Not applicable.

Halogen content: Not applicable.

Immersion test: Not applicable.

Smoke index: Not applicable.

Toxicity index: Not applicable.

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Durometer hardness: Not applicable.

Weathering: Not applicable.

Abrasion resistance: Not applicable.

Tear strength: Not applicable.

Heat distortion: Not applicable.

Physical tests on unaged jacket: Not applicable.

Physical tests on aged jacket: Not applicable.

Hot oil immersion: Not applicable.

Weight: 1.5 pounds per 100 feet maximum.

Electrical:

Test frequency:

50 MHz to 18 GHz, M17/205-00018
50 MHz to 50 GHz, M17/205-00050

Spark test: 2,000 V rms, + 25%, -0%.

Voltage withstanding: 3,000 V rms, minimum.

Corona extinction voltage: 1.9 kV rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

Structural return loss: 1.15:1 at 12 GHz, 1.25:1 at 18 GHz, 1.30:1 at 26.5 GHz, 1.40:1 at 50 GHz, maximum.

Capacitance: 27 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

Part or Identifying Number (PIN): M17/205-00018 and M17/205-00050.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, ME, MI
Navy - AS, MC, OS, SH
Air Force - 19, 80, 99
DLA - ES, IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-2016-01)

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